

4-Bit Binary Arithmetic Calculator

Design & Simulation Using Combinational Logic

ABSTRACT

This project presents the design and simulation of a 4-bit binary arithmetic calculator built entirely from combinational logic circuits. The system performs both addition and subtraction on 4-bit operands using full-adder chains and two's-complement techniques, with logic gates arranged to compute sum, difference, carry, and overflow outputs. Emphasis is placed on gate-level design, Boolean optimization, and functional verification through simulation waveforms. The work highlights how such combinational structures form the foundational building blocks of Arithmetic Logic Units (ALUs) at the heart of processors and modern digital computing systems.

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